

NetVault-Tactical

TACTICAL NETWORK CONFIGURATION BACKUP & RECOVERY SYSTEM

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1. Introduction

1.1 Purpose of This Manual

This User Manual provides complete instructions for the installation, configuration, operation, and maintenance of NetVault-Tactical (Product Code: NVT-1.8). It is intended for network engineers, signal operators, and system administrators responsible for managing tactical network infrastructure in forward-deployed environments. Readers are assumed to have a working knowledge of TCP/IP networking, Linux command-line operations, and basic network device administration.

1.2 Scope

This manual covers all user-facing functionality of NetVault-Tactical Version 1.8, including system setup, node registration, backup scheduling, configuration restoration, cloud synchronization management, and routine maintenance. It does not cover advanced developer integration or classified deployment procedures, which are addressed in separate technical documentation.

1.3 How to Use This Manual

This manual is organized as follows:

- Sections 1–2 provide introduction and safety information — read before first use.
- Section 3 covers technical specifications and system requirements.
- Sections 4–5 describe installation and initial configuration.
- Section 6 details daily operational procedures.
- Section 7 explains the restoration workflow.
- Section 8 covers troubleshooting and error codes.

- Section 9 provides maintenance guidelines.

NOTE

Steps marked with [REQUIRED] must be completed in order. Steps marked [OPTIONAL] may be skipped based on operational needs. Always read the General Safety section (Section 2) before proceeding.

2. General Safety Prescriptions

2.1 Cybersecurity Safety

WARNING

NetVault-Tactical stores encrypted network configuration files that may contain sensitive topology information. Unauthorized access to backup data could enable adversarial network reconnaissance. All installations **MUST** enforce zero-trust authentication via ICAM integration. Never disable TPM-backed encryption without authorization.

- Always use strong, unique service account credentials for the application daemon.
- Rotate encryption keys at intervals not to exceed 90 days in operational deployments.
- Enable Secure Boot on all host servers before deploying NetVault-Tactical.
- Restrict SSH access to the management interface to trusted subnets only.
- Audit access logs weekly to detect unauthorized access attempts.

2.2 Physical Security

- Servers running NetVault-Tactical must be physically secured in locked enclosures.
- Enable chassis intrusion detection on all edge servers.
- Apply tamper-evident seals to server access panels in unsecured forward locations.
- Uninterruptible power supplies (UPS) must be tested monthly to ensure runtime integrity.

2.3 Data Handling

- Backup archives must not be copied to unencrypted removable media.
- Before decommissioning a server, perform a NIST 800-88-compliant drive sanitization.
- Cloud synchronization must only target pre-approved, accredited cloud tenants (e.g., JWCC).
- All configuration data in transit must use TLS 1.3 or higher.

2.4 Operational Safety

CAUTION

NetVault-Tactical backup processes consume network bandwidth. Cloud synchronization is automatically throttled during active engagement periods, but operators must verify that bandwidth management settings are correct for their environment before deployment.

- Do not perform restoration operations on live, mission-critical nodes without a maintenance window.
- Always verify the integrity hash of a backup archive before initiating restoration.
- Test restoration procedures in a lab environment before first operational use.

3. Technical Specifications

3.1 System Requirements

Component	Specification
Processor	Intel Core i7 (12th Gen+) or AMD Ryzen 7 / Server-grade equivalent
RAM (Minimum)	16 GB DDR4 (prototype/testing)
RAM (Recommended)	32 GB DDR4-3200 ECC (production)
RAM (Large Site)	64 GB DDR4 ECC (50+ nodes)
Primary Storage	2x 1 TB NVMe SSD, RAID-1 mirrored
Cold Storage	4 TB HDD (optional overflow archive)
Network	Dual Gigabit Ethernet NIC (management + data separation)
Security Module	TPM 2.0 (required for at-rest encryption)
Power	Redundant PSU + UPS (30-min minimum runtime)
Form Factor	Rack-mounted or compact; MIL-STD-810 compliant preferred

3.2 Software Requirements

Component	Specification
Operating System	Ubuntu Server 22.04 LTS or RHEL 9 (STIG-hardened)
Kernel	Linux 5.15 or higher
Container Runtime	Docker Engine 24.x or newer
Orchestration	K3s (Kubernetes lightweight) — optional
Python Runtime	Python 3.10+
Encryption Library	OpenSSL 3.0+
SNMP Toolkit	net-snmp 5.9+
SSH Library	Paramiko 3.x / OpenSSH 8.x+

Database	PostgreSQL 15+ (local config registry)
Web Interface	Nginx 1.24+ (reverse proxy for management UI)

3.3 Supported Device Types

- Cisco IOS/IOS-XE/NX-OS routers and switches
- Juniper Junos-based devices
- Palo Alto Networks firewalls (via API)
- SATCOM terminal controllers (via SSH/SNMP)
- Generic SSH/SNMP-accessible network devices

4. Relation to Other Documents

NetVault-Tactical is designed to complement and integrate with existing DoD network management and security frameworks. The following documents are directly relevant to the deployment and operation of this product:

Document	Relationship
NIST SP 800-111	Governs storage encryption standards applied to backup archives
NIST SP 800-53 Rev 5	Security control baseline for system hardening and access control
DoD STIG for RHEL 9	Mandatory OS hardening guide for production RHEL deployments
JWCC Cloud Tenant Policy	Cloud storage tenant requirements and accreditation procedures
MIL-STD-810H	Environmental testing standard for hardware platform selection
Zero Trust Reference Architecture (DoD)	Authentication and identity management design framework
Network Infrastructure STIG	Device configuration baseline for registered network nodes

5. Product Description and Composition

5.1 Overview

NetVault-Tactical is a software application designed to automate the secure backup, local storage, and cloud synchronization of configuration files for forward-deployed tactical network terminals and

equipment. Its mission is to ensure rapid network reconstitution following hardware failure, cyber-attack, or electronic warfare interference in disconnected, intermittent, and limited (DIL) bandwidth environments.

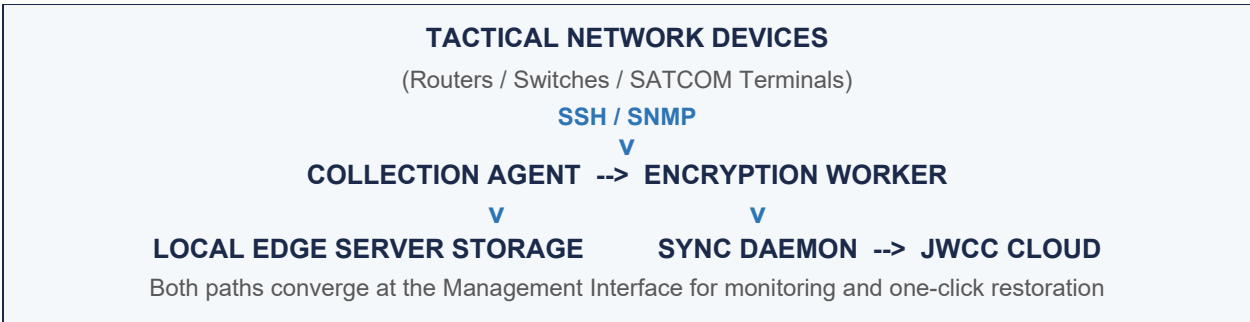
5.2 System Architecture

The application is composed of five primary service components, each containerized and independently deployable:

Component	Function
Scheduler Engine	Manages backup job queues, schedules, and event triggers across all registered nodes
Collection Agent	Connects to network devices via SSH/SNMP; extracts and normalizes configuration data
Encryption Worker	Applies AES-256-GCM encryption to all backup archives; manages TPM-backed key storage
Sync Daemon	Monitors connectivity and pushes encrypted archives to cloud (JWCC) when bandwidth permits; enforces throttle policies
Management Interface	Web-based dashboard for node registration, job monitoring, restoration, and audit log review

5.3 Data Flow Summary

Figure 1 — System Architecture Overview (Conceptual)



5.4 Icon Reference — Dashboard Status Icons

Icon Label	Meaning
[GREEN] Node Online	Device is reachable; last backup completed successfully
[YELLOW] Backup Pending	Backup job scheduled and queued; awaiting execution
[ORANGE] Sync Throttled	Cloud sync active but rate-limited; bandwidth conservation mode

[RED] Node Unreachable	Device is offline or SSH/SNMP connection failed
[RED] Backup Failed	Last backup attempt encountered an error; see audit log
[BLUE] Restoring	Configuration restoration operation in progress
[GRAY] Node Disabled	Node is registered but backup is manually paused
[LOCK] Encrypted	Archive is encrypted and integrity-verified

6. Installation and Initial Configuration

6.1 Pre-Installation Checklist

- Host server meets or exceeds minimum hardware specifications (Section 3.1).
- Ubuntu Server 22.04 LTS or RHEL 9 installed and updated.
- Docker Engine 24.x installed and running.
- TPM 2.0 enabled in BIOS/UEFI and accessible to the OS.
- ICAM service account credentials provisioned.
- Network firewall rules allow SSH (port 22) and SNMP (UDP 161) to/from managed devices.
- Cloud tenant endpoint and credentials obtained from JWCC administrator.

6.2 Installation Steps

1. [REQUIRED] Pull the NetVault-Tactical container image:
`docker pull registry.netvault.mil/netvault-tactical:1.8`
2. [REQUIRED] Initialize the local database and key store:
`docker run --rm netvault-tactical:1.0 nvt-init --tpm-bind`
3. [REQUIRED] Start the application stack:
`docker compose -f /opt/netvault/docker-compose.yml up -d`
4. [REQUIRED] Verify all services are running:
`docker compose ps`
5. [REQUIRED] Access the management interface at `https://<server-ip>:8443`
6. [REQUIRED] Complete first-login setup wizard (admin credentials, ICAM binding, cloud endpoint).

6.3 Registering Network Nodes

To add a network device to NetVault-Tactical for backup management:

Field	Description
Node Name	Friendly display name (e.g., RTR-FOB-ALPHA-01)
IP Address	Management IP of the target device
Device Type	Cisco IOS / Junos / PAN-OS / Generic SSH / SNMP

Auth Method	SSH key (recommended) or SSH password
SNMP Community	SNMPv3 credentials or community string
Backup Schedule	Daily / Event-Driven / Manual (see Section 6.4)
Priority Tier	1 (Critical) / 2 (Important) / 3 (Standard)

6.4 Configuring Backup Schedules

Schedule Mode	Behavior
Daily (Scheduled)	Runs at a configured time each day; default 02:00 local server time
Event-Driven	Triggers on detected SNMP change notification (config-change trap)
Interval	Runs every N hours; configurable from 1–24 hours
Manual	Backup only when operator initiates from dashboard or CLI

7. Configuration Restoration Procedure

WARNING

Restoration overwrites the current running configuration on the target device. Always confirm the target device is a replacement unit or that the current configuration can be discarded before proceeding. Restoration of a live, in-use node may cause a brief service interruption.

7.1 Standard Restoration (Dashboard)

7. Log in to the NetVault-Tactical Management Interface.
8. Navigate to Nodes > [Node Name] > Backup Archive.
9. Select the desired backup version from the archive list (most recent is default).
10. Click Verify Archive — confirm the integrity hash matches.
11. Click Restore Configuration.
12. Confirm the restoration target and accept the confirmation prompt.
13. Monitor progress in the Restoration Status panel; node icon will display [BLUE] Restoring.
14. Upon completion, verify device connectivity and configuration correctness.

7.2 CLI Restoration (Offline / Emergency)

If the management interface is unavailable, restoration can be performed via CLI:

```
nvt-restore --node <node-name> --version latest --target <device-ip>
```

To list available versions for a node:

```
nvt-list-archives --node <node-name>
```

7.3 Restoration from Local vs. Cloud Archive

- NetVault-Tactical will first attempt restoration from the local edge server archive.
- If the local archive is unavailable or corrupted, it will automatically fall back to the cloud archive.
- Manual cloud-only restoration can be forced with the `--source cloud` flag.

8. Troubleshooting and Error Codes

8.1 Common Error Codes

Error Code	Description and Remediation
NVT-001	Node Unreachable — Verify device IP, SSH credentials, and firewall rules.
NVT-002	Authentication Failed — Confirm SSH key or password; rotate credentials if needed.
NVT-003	SNMP Timeout — Verify SNMPv3 credentials and UDP port 161 is open.
NVT-004	Backup Archive Corrupt — Delete and re-run backup; check disk health.
NVT-005	Encryption Key Error — Re-bind TPM key store: <code>nvt-keytool --rebind</code> .
NVT-006	Cloud Sync Failure — Verify JWCC endpoint URL, credentials, and WAN connectivity.
NVT-007	Restoration Hash Mismatch — Archive integrity compromised; restore from alternate version.
NVT-008	Storage Capacity Warning — Local storage is above 85%; purge old archives or expand storage.
NVT-009	Scheduler Engine Down — Restart container: <code>docker restart nvt-scheduler</code> .
NVT-010	ICAM Auth Rejected — Service account token expired; re-authenticate via ICAM portal.

8.2 Viewing System Logs

All application logs are accessible via the dashboard under System > Audit Logs, or via CLI:

```
docker logs nvt-scheduler --tail 100
docker logs nvt-collection-agent --tail 100
docker logs nvt-sync-daemon --tail 100
```

9. Maintenance Guidelines

9.1 Routine Maintenance Schedule

Frequency	Task
Daily	Review dashboard for failed backups; resolve any NVT error codes.
Weekly	Review audit logs for unauthorized access attempts.
Monthly	Test UPS runtime and redundant PSU failover. Test restoration procedure in lab environment.
90 Days	Rotate encryption keys using <code>nvt-keytool --rotate</code> . Update service account credentials.
As Needed	Purge archives older than retention policy (default: 90 days). Apply OS and container security patches.

9.2 Archive Retention Management

NetVault-Tactical retains backup archives according to a configurable retention policy. The default policy retains the last 30 daily backups and all event-driven backups for 90 days. To modify the retention policy, navigate to Settings > Backup Policy > Retention in the dashboard.

9.3 Software Updates

- Update notifications are displayed in the dashboard banner when new versions are available.
- To update, pull the new container image and re-deploy using the docker compose workflow.
- Always review the release notes before updating in a production environment.
- Database migrations are handled automatically during container startup.

NOTE

NetVault-Tactical does not require internet access to function. All update distribution in air-gapped environments must be performed via approved media transfer procedures consistent with local security policies.

10. Glossary

Term	Definition
AES-256-GCM	Advanced Encryption Standard, 256-bit key, Galois/Counter Mode — the encryption algorithm used for all backup archives.
DIL	Disconnected, Intermittent, and Limited — describes bandwidth-constrained tactical network environments.
ICAM	Identity, Credential, and Access Management — the DoD framework for zero-trust authentication.
JWCC	Joint Warfighting Cloud Capability — the DoD enterprise cloud contract and infrastructure.
RAID-1	Redundant Array of Independent Disks, Level 1 — mirroring configuration that duplicates data across two drives.
SNMP	Simple Network Management Protocol — used to collect configuration and status data from network devices.
SSH	Secure Shell — encrypted protocol used for remote device access and configuration extraction.
STIG	Security Technical Implementation Guide — DoD cybersecurity hardening standards.
TPM 2.0	Trusted Platform Module — hardware chip providing secure key storage for encryption operations.
Zero Trust	Security model requiring continuous verification of every user and device, with no implicit trust.